

What Factors Affect The World Happiness Report the Most

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I. Abstract

The World Happiness Report ranks countries based on self-reported happiness surveys. It also explores the connection between happiness and various life factors, including economic ones. This report allows researchers to analyze how income, inequality, social safety nets, and trust institutions influence national well-being across the globe.

II. Introduction and Summary

Happiness and well-being are more than just fleeting positive emotions. They represent a state of overall satisfaction with life, impacting everything from individual health and productivity to social cohesion and economic growth. Recognizing this critical role, the United Nations launched the World Happiness Report (WHR) in 2012. This annual report utilizes Gallup World Poll data to rank countries based on self-reported happiness surveys. Additionally, the WHR delves deeper, exploring the connection between happiness and various life factors, including economic ones. This paper utilizes the rich trove of data provided by the WHR to shed light on the economic forces shaping happiness across nations. Through the lens of econometrics, we aim to analyze how key economic factors like income, income inequality, social safety nets, and the strength of trust institutions influence national well-being. By employing econometric methods, we can move beyond simple correlations and establish more robust evidence for the causal relationships between these economic factors and happiness.

Income has traditionally been considered a driver of happiness. However, theories like the Easterlin Paradox (Easterlin, 1974) posit a limited impact of income beyond a certain point. Social comparison theory (Fehr & Tyran, 2001) suggests relative income might be more important. Additionally, social spending on education and healthcare is crucial for human capital, health outcomes, and potentially greater happiness.

Institutional quality also shapes happiness. Studies suggest high levels of political trust (Freitag & Leigh, 2015) create security and predictability, leading to greater happiness. Robust social safety nets provide a safety cushion during hardship (Helliwell & Putnam, 2007). Finally, measuring a higher rate of political and economic freedoms have been linked to higher happiness levels across countries.

III. Model Description

This project, utilizing the dataset WHR_2023.csv and employing econometric techniques, aims to contribute to a deeper understanding of how economic policies and institutions influence national happiness. We will estimate a linear regression model to analyze the relationship between happiness and various economic and institutional factors. The general form of the model is:

$$\text{happiness_score} = \beta_0 + \beta_1 \times \text{gdp_per_capita} + \beta_2 \times \text{social_support} + \beta_3 \times \text{healthy_life_expectancy} + \beta_4 \times \text{freedom_to_make_life_choices} + \beta_5 \times \text{generosity} + \beta_6 \times \text{perceptions_of_corruption} + \epsilon$$

Here are the variables with a small description so we know what we are working with:

β 1: GDP per Capita, A higher GDP per capita is expected to positively influence happiness scores. Economic prosperity provides individuals with better access to resources and opportunities, contributing to overall life satisfaction and well-being.

β 2. Social Support: Countries with strong social support systems, including family, friends, and community networks, are likely to have higher happiness scores. Social connections and support networks play a crucial role in buffering against stress and adversity, enhancing subjective well-being.

β 3. Healthy Life Expectancy: Longer life expectancy and better health outcomes are associated with higher happiness levels. Access to quality healthcare, healthy lifestyle choices, and overall well-being contribute positively to individuals' perception of happiness and life satisfaction.

β 4. Freedom to Make Life Choices: Societies that promote individual freedoms and autonomy tend to have higher happiness scores. The ability to make personal choices regarding one's life, career, and relationships fosters a sense of empowerment and fulfillment.

*β*₅. Generosity: Generosity, reflected in charitable giving and altruistic behavior, is expected to positively influence happiness scores. Acts of kindness and generosity contribute to social cohesion and emotional well-being, enhancing overall happiness levels.

*β*₆. Perceptions of Corruption: Countries with lower levels of perceived corruption are likely to have higher happiness scores. Corruption undermines trust in institutions and societal cohesion, leading to lower subjective well-being among citizens.

Overall, each value should be positively correlated with the dependent variable, happiness score. It's also worth mentioning that each variable has been standardized meaning each variable has been "scaled" by population, making it easier to compare.

IV. Data

Diving into the data, we are going to look at the basic summary statistics of our model. Then we are going to analyze the OLS model by looking at parameter estimates, standard errors, T and F stats, and look at the overall model fit. Then we are going to look for autocorrelation, heteroskedasticity, and multicollinearity.

Summary Statistics

First, let's look at our OLS model results:

count	Variable	Mean	Variance	Minimum	25%	Median	75%	Maximum
137.0	Happiness Score	5.539795620437956	1.139928746087732	1.859	4.724	5.684	6.334	7.804
137.0	GDP per capita	1.406985401459854	0.4329626730515043	0.0	1.099	1.449	1.798	2.2
137.0	Social Support	1.1562116788321168	0.3263224999601606	0.0	0.962	1.227	1.401	1.62
136.0	Health Life Expectancy	0.36617647058823527	0.15669110364275413	0.0	0.2485	0.3895	0.4875	0.702
137.0	Freedom of choice	0.54	0.1495013279500337	0.0	0.455	0.557	0.656	0.772
137.0	Generosity	0.14847445255474453	0.07605349126572332	0.0	0.097	0.137	0.199	0.422
137.0	Perception of corruption	0.1458978102189781	0.1267229658297935	0.0	0.06	0.111	0.187	0.561

First, looking at R^2 at 0.828 which indicates that the model explains a high proportion of the variance in happiness scores. Also, the adjusted R^2 at 0.820 is not overfitting the data by including any unnecessary variables. Now we are going to interpret the variables by their P-values and how they fit to the model.

The coefficient estimates provide insights into the direction and magnitude of the relationship between each independent variable and happiness scores. Notably, variables such as GDP per capita, social support, and freedom to make life choices exhibit statistically significant positive effects on happiness scores. In contrast, the impact of healthy life expectancy, generosity, and perceptions of corruption appears to be less significant.

Social Support and Freedom to Make Life Choices: Both variables have positive and statistically significant coefficients (p-value < 0.0001), suggesting that stronger social networks and greater autonomy in life decisions are associated with higher happiness scores.

Perceptions of Corruption: The negative and statistically significant coefficient (p-value = 0.0052) indicates that lower levels of perceived corruption are linked to increased happiness.

GDP per Capita: The positive and significant coefficient (p-value = 0.0051) suggests that higher national economic prosperity is associated with greater happiness scores.

Healthy Life Expectancy and Generosity: These variables do not exhibit statistically significant effects on happiness scores at the conventional 5% level (p-value = 0.1557 and 0.6523, respectively). This may warrant further investigation or exploration of alternative model specifications.

Based on these results alone, we can determine that social support, freedom to make life choices, perceptions of corruption, and GDP per capita are significant determinants of happiness scores based on our OLS regression analysis. Individuals with stronger social networks, greater autonomy, and residing in societies with lower perceived corruption and higher economic prosperity tend to report higher levels of happiness.

Finally, we want to test our model to see if there is any autocorrelation, heteroskedasticity, and multicollinearity. The Durbin-Watson statistic of 1.841 suggests that there may be some positive autocorrelation present in the residuals.

The Breusch-Pagan test is used to assess whether there is heteroskedasticity in the regression residuals. Here are the results of the Breusch-Pagan test for heteroskedasticity:

Test Statistic (LM): 13.689783
Degrees of Freedom: 6
p-value: 0.0333

The null hypothesis of the Breusch-Pagan test is that there is homoscedasticity (constant variance of residuals) in the regression model. Since the p-value of the test (0.0333) is less than the significance level of 0.05, we reject the null hypothesis. This indicates that there is evidence of heteroskedasticity in the regression residuals.

Next, based on the VIF values, there does not appear to be any significant multicollinearity problem among the independent variables in the regression model.

Social Support: VIF = 3.011

Healthy Life Expectancy: VIF = 3.732

Freedom to Make Life Choices: VIF = 1.591

Generosity: VIF = 1.193

Perceptions of Corruption: VIF = 1.425

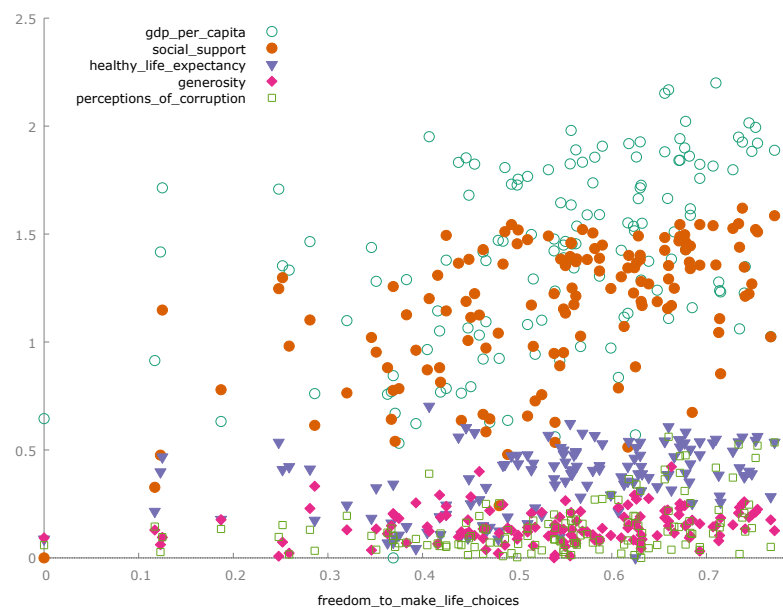
GDP per Capita: VIF = 4.268

Although GDP per Capita has a higher VIF than the other values, it doesn't seem to have any severe correlation relative to the other independent variables.

V. Conclusion

This paper employed data provided by the WHR to analyze how economic policies and institutions influence national happiness. We used an econometric OLS model to examine the relationship between happiness and various economic and social factors. Our findings shed light on these key influences. Social connections and a sense of control over life choices are crucial for happiness. Countries with strong social support systems, including family, friends, and community networks, tend to have higher happiness scores. Similarly, societies that promote individual freedoms and autonomy in life decisions are associated with greater happiness. Economic prosperity also plays a role. Our analysis suggests that higher GDP per capita is positively linked to happiness scores. This indicates that economic resources contribute to a better standard of living and potentially greater life satisfaction. However, the relationship might not be linear. Future research could explore if the impact of economic growth on happiness diminishes as wealth increases. Trust in institutions also matters. Lower levels of perceived corruption are linked to increased happiness. This suggests that a fair societal environment and

trust in institutions contribute to subjective well-being. The influence of health and generosity on happiness appears more nuanced. While included in the model, they did not exhibit statistically significant effects. Further investigation or alternative models might be needed to understand their potential role.



This graph shows the GDP per capita and social support have the greatest impact on happiness levels, while generosity has the lowest impact. Note that This study acknowledges limitations, including that this may not purely represent people's happiness and there's an inability to establish causal relationships due to the cross-sectional nature of the data. Although it's difficult to quantify happiness, these results can help provide information for government bodies to help make decisions on public policy and help increase satisfaction levels across the globe.

VI. Appendix

OLS Model:

Model 1: OLS, using observations 1-137 (n = 136)

Missing or incomplete observations dropped: 1

Dependent variable: happiness_score

	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-ratio</i>	<i>p-value</i>	
const	1.52863	0.202286	7.557	<0.0001	***
social_support	1.55459	0.221249	7.026	<0.0001	***
healthy_life_expec tancy	0.734729	0.514450	1.428	0.1557	
freedom_to_make_ life_choices	1.75686	0.351674	4.996	<0.0001	***
generosity	0.270823	0.599690	0.4516	0.6523	
perceptions_of_cor ruption	1.11454	0.392130	2.842	0.0052	***
gdp_per_capita	0.565787	0.198639	2.848	0.0051	***
Mean dependent var	5.544441	S.D. dependent var		1.142841	
Sum squared resid	30.32009	S.E. of regression		0.484809	
R-squared	0.828041	Adjusted R-squared		0.820043	
F(6, 129)	103.5297	P-value(F)		7.27e-47	
Log-likelihood	-90.91822	Akaike criterion		195.8364	
Schwarz criterion	216.2250	Hannan-Quinn		204.1218	

Breusch-Pagan

Breusch-Pagan test for heteroskedasticity

OLS, using observations 1-137 (n = 136)

Missing or incomplete observations dropped: 1

Dependent variable: scaled uhat^2 (Koenker robust variant)

	coefficient	std. error	t-ratio	p-value	
const	0.434493	0.147246	2.951	0.0038	***

social_support	-0.189575	0.161049	-1.177	0.2413
healthy_life_exp~	-0.424464	0.374472	-1.134	0.2591
freedom_to_make_~	-0.114375	0.255986	-0.4468	0.6558
generosity	-0.453108	0.436518	-1.038	0.3012
perceptions_of_c~	0.195813	0.285434	0.6860	0.4939
gdp_per_capita	0.0288511	0.144590	0.1995	0.8422

Explained sum of squares = 1.79811

Test statistic: LM = 13.689783,
with p-value = P(Chi-square(6) > 13.689783) = 0.033300

Variance Inflation Factors:

Minimum possible value = 1.0

Values > 10.0 may indicate a collinearity problem

social_support	3.011
healthy_life_expectancy	3.732
freedom_to_make_life_choices	1.591
generosity	1.193
perceptions_of_corruption	1.425
gdp_per_capita	4.268

$VIF(j) = 1/(1 - R(j)^2)$, where $R(j)$ is the multiple correlation coefficient
between variable j and the other independent variables

Belsley-Kuh-Welsch collinearity diagnostics:

variance proportions

lambda	cond	const	social_s~	healthy_~	freedom_~	generosi~	percepti~	gdp_per_~
6.296	1.000	0.001	0.001	0.001	0.001	0.004	0.005	0.000
0.334	4.339	0.007	0.001	0.000	0.002	0.062	0.706	0.000
0.238	5.146	0.000	0.004	0.031	0.000	0.431	0.105	0.008
0.061	10.120	0.156	0.001	0.220	0.202	0.407	0.025	0.003
0.035	13.467	0.554	0.019	0.004	0.596	0.006	0.036	0.025
0.021	17.438	0.082	0.704	0.502	0.168	0.006	0.044	0.030
0.014	20.888	0.199	0.269	0.241	0.031	0.085	0.079	0.933

lambda = eigenvalues of inverse covariance matrix (smallest is 0.014431)

cond = condition index

note: variance proportions columns sum to 1.0

According to BKW, cond ≥ 30 indicates "strong" near linear dependence, and cond between 10 and 30 "moderately strong". Parameter estimates whose variance is mostly associated with problematic cond values may themselves be considered problematic.

Count of condition indices ≥ 30 : 0

Count of condition indices ≥ 10 : 4

Variance proportions ≥ 0.5 associated with cond ≥ 10 :

const	social_s~	healthy_~	freedom_~	generosi~	gdp_per_~
0.992	0.993	0.967	0.997	0.503	0.991

References

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